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Interface morphogenesis with a deformable secondary phase in solid-state lithium batteries

Sun Geun Yoon[1], Bairav S. Vishnugopi[2], Douglas Lars Nelson[3], Adrian Xiao Bin Yong[4,5], Yingjin Wang[6], Stephanie Elizabeth Sandoval[1,3], Talia A. Thomas[1], Kelsey Anne Cavallaro[3], Pavel Shevchenko[7], Elif Pınar Alsaç[1], Congcheng Wang[1], Aditya Singla[2], Julia R. Greer[6], Elif Ertekin[5,8], Partha P. Mukherjee[2], Matthew T. McDowell[1,3]*

The complex morphological evolution of lithium metal at the solid-state electrolyte interface limits performance of solid-state batteries, leading to inhomogeneous reactions and contact loss. Inspired by biological morphogenesis, we developed an interfacial self-regulation concept in which a deformable secondary phase dynamically aggregates at the interface in response to local electro-chemo-mechanical stimuli, enhancing contact. The stripping of a lithium electrode that contains 5 to 20 mole % electrochemically inactive sodium domains causes spontaneous sodium accumulation across the interface, with the sodium deforming to attain intimate electrical contact without blocking lithium transport. This process, characterized with operando x-ray tomography and electron microscopy, mitigates voiding and improves cycling at low stack pressures. The counterintuitive strategy of adding electrochemically inactive alkali metal to improve performance demonstrates the utility of interfacial self-regulation for solid-state batteries.

The high specific capacity of lithium (Li) metal and its compatibility with various solid-state electrolytes (SSEs) has focused efforts on developing Li metal solid-state batteries (SSBs) (*1*–*3*). However, a critical limitation of Li metal anodes for SSBs is the loss of contact at the SSE interface during Li stripping (battery discharge) owing to void formation (*1*, *2*, *4*). The formation of voids is detrimental because they cause high interfacial impedance and current focusing during charging, resulting in Li filament growth and short circuiting (*5*–*8*).

The mechanisms that govern void formation are complex, and the process is affected by current density, stack pressure, chemo-mechanics, and other factors (*5*, *7*, *9*–*13*). During stripping, vacancies are formed in the Li metal because of Li^+ transfer into the SSE. Because Li metal exhibits relatively low self-diffusivity (~10^{-10} cm^2 s^{-1}) (*14*–*16*), the vacancies tend to accumulate to nucleate voids (Fig. 1A). Higher current densities increase the vacancy formation rate and accelerate void formation (*7*, *11*, *14*). The application of stack pressure can deform soft Li metal (bulk yield strength of 0.8 MPa) to promote contact retention during stripping, but stack pressures greatly exceeding the yield strength have been shown to be required at moderate current densities to prevent void formation (>7 MPa for 1 mA cm^{-2}) (*5*, *7*). Moreover, Li shows heightened yield strength at submicrometer length scales, which may result in needing even higher stack pressures at SSE interfaces with nanoscale roughness (*17*–*19*). Critically, stack pressures must be low for achieving high cell-level specific energy (fig. S1), ideally being less than ~1 MPa for practical applications (*17*, *20*–*22*).

Interlayers comprising various materials, such as carbon (C) or Li alloys, have been shown to enable improved SSB performance. In general, these interlayer materials enhance Li transport toward the SSE interface, homogenize current distributions, and can help maintain interfacial contact (*23*–*26*). However, these approaches have primarily used hard or brittle interlayer materials to passively control Li transport, and some of these materials undergo transformations themselves (*27*–*29*). Relatively high stack pressures are therefore usually required to maintain interfacial contact (fig. S2). Interfacial strategies are needed that feature materials that can dynamically respond to local driving forces to promote contact retention even at low stack pressures.

We developed an approach to regulate SSB interfaces that is inspired by biological morphogenesis, which is the process by which living cells self-organize to form patterns and structures through multi–length scale chemo-mechanical interactions (*30*, *31*). Our approach involves self-regulated accumulation of a secondary phase at the SSE interface in response to the evolving morphological irregularities caused by Li stripping, sustaining interfacial contact and improving behavior at low stack pressures. The approach is demonstrated by use of Li electrodes that contain distributed metallic sodium (Na) domains with <20% Na. Na is electrochemically inactive, electrically conductive, and mechanically softer than Li (the bulk yield strength of Na is 0.16 to 0.24 MPa) (table S3). Na dynamically accumulates at the interface during Li stripping and plastically deforms to create intimate electrical contact (Fig. 1B), which suppresses void formation and enhances stripping capacity at stack pressures <1 MPa without blocking Li. The accumulation and deformation of the Na enables morphogenic self-regulation at the Li-SSE interface, with Na contact spontaneously emerging without external guidance in response to electrochemical and mechanical stimuli (*30*, *31*). The electrical conductivity of Na facilitates subsequent uniform Li plating, avoiding current constrictions and enabling stable cycling at low stack pressures.

Electrochemical stripping and plating

We fabricated Li-Na foils containing between 2.5 and 20 mol % Na using accumulative roll bonding (supplementary materials, materials and methods). In fig. S3, we show cross-sectional cryogenic focused-ion beam (cryo-FIB) scanning electron microscopy (SEM) images of the binary metals, in which Na domains a few to 10 μm in size are surrounded by a continuous Li matrix. Na and Li are immiscible at room temperature, giving rise to the two-phase microstructure (*32*). The electrode potential of Na is greater than that of Li (–2.71 and –3.04 V versus the standard hydrogen electrode, respectively), suggesting that Na should remain metallic and electrochemically inactive during Li redox processes (*33*).

In Fig. 1C, we show voltage curves during electrochemical stripping of Li from various Li-Na electrodes at stack pressures between 0 and 3.2 MPa. Solid-state half cells were used with argyrodite Li_6PS_5Cl SSE (LPSC). Pure Li showed almost 30 mA·hour cm^{-2} stripping capacity up to a 0.5 V cutoff at 3.2 MPa stack pressure, but substantially less Li was accessible at lower stack pressures (2.2 mA·hour cm^{-2} at 0.8 MPa and 0.4 mA·hour cm^{-2} at 0 MPa). This is a result of uncompensated void formation during stripping at low stack pressure (*23*, *34*). The stripped capacities of the Li-Na electrodes were generally higher than pure Li at stack pressures below 1.6 MPa. For example, the 5% Na electrode delivered 16 mA·hour cm^{-2} at 0.8 MPa stack pressure, and the 20% Na electrode was able to strip 5.3 mA·hour cm^{-2} at 0 MPa. The voltage profiles of the electrodes shown in Fig. 1C had voltage variations at 0.15 to 0.25 V (fig. S4), which were absent during stripping of pure Li electrodes. This behavior likely arises from some

[1]George W. Woodruff School of Mechanical Engineering, Georgia Institute of Technology, Atlanta, GA, USA. [2]School of Mechanical Engineering, Purdue University, West Lafayette, IN, USA. [3]School of Materials Science and Engineering, Georgia Institute of Technology, Atlanta, GA, USA. [4]Department of Materials Science and Engineering, University of Illinois Urbana-Champaign, Urbana, IL, USA. [5]Materials Research Laboratory, University of Illinois Urbana-Champaign, Urbana, IL, USA. [6]Division of Engineering and Applied Sciences, California Institute of Technology, Pasadena, CA, USA. [7]Advanced Photon Source, Argonne National Laboratory, Lemont, IL, USA. [8]Department of Mechanical Science and Engineering, University of Illinois Urbana-Champaign, Urbana, IL, USA. *Corresponding author. Email: mattmcdowell@gatech.edu

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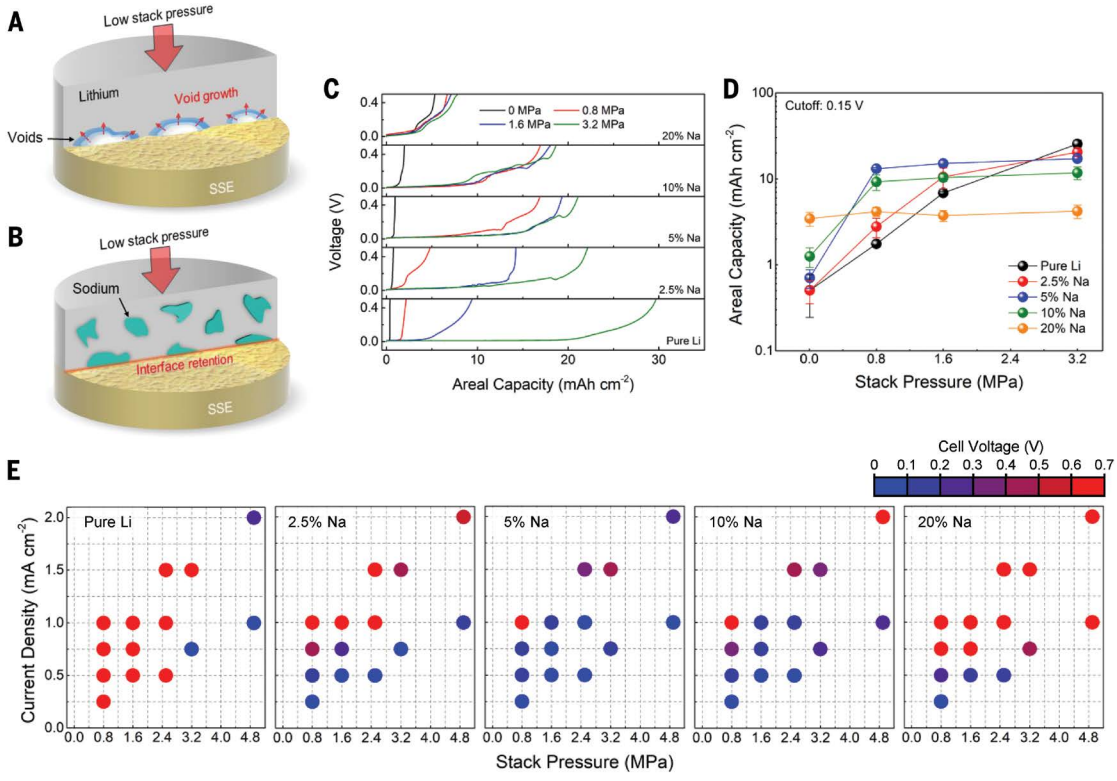


Fig. 1. Electrochemical stripping and plating tests of Li-Na electrodes. (**A** and **B**) Schematics of (A) void formation during pure Li stripping at low stack pressure and (B) Na accumulation and void mitigation during stripping of a Li-Na electrode. (**C**) Galvanostatic voltage profiles during Li stripping from various Li-Na electrodes at different stack pressures. The cells used Li_6PS_5Cl SSE, Li metal counter electrodes, and a current density of 0.25 mA cm^{-2}. (**D**) Average stripped areal capacity from different Li-Na electrodes (three experiments each) as a function of stack pressure. The stripped capacities were measured at a cutoff voltage of 0.15 V. Error bars indicate standard deviation. (**E**) Color maps showing the cell voltage after 3 mA·hour cm^{-2} of stripping or at a cutoff voltage of 0.62 V if this capacity was not reached, as a function of stack pressure (0.8 to 4.8 MPa, *x* axis) and current density (0.25 to 2.0 mA cm^{-2}, *y* axis). Working electrodes comprising pure Li, 2.5% Na, 5% Na, 10% Na, and 20% Na were used (left to right, respectively). A Li–In alloy (LiIn–In, +0.62 V versus Li/Li^+) electrode was used as the counter electrode to avoid short circuiting. All percentages are molar percentages.

interfacial oxidation of the Na because the potential is near the Na electrode potential.

We show in Fig. 1D statistical trends of the average stripped Li capacities for the different electrodes at varying stack pressure. Whereas the pure Li and 2.5% Na electrodes showed monotonically increasing capacity with increasing stack pressure, the other composite electrodes showed similar capacities at stack pressures between 0.8 and 3.2 MPa. The electrode with 20% Na is a limiting example, showing approximately invariant stripped capacity regardless of stack pressure, with lower capacities at higher stack pressures that may arise from early blocking of Li transport to the interface.

A comprehensive summary of the initial stripping behavior of these electrodes at various current densities and stack pressures is shown in the maps in Fig. 1E. The color of the points indicates the voltage either after 3.0 mA·hour cm^{-2} was stripped or at the cutoff voltage of 0.62 V if this capacity was not reached (fig. S5A). The pure Li electrode can only strip this capacity at stack pressures >3.2 MPa. By contrast, the electrodes with 5 and 10% Na allowed for successful stripping with <0.1 V at stack pressures down to 0.8 MPa and varying current densities. The stable stripping region for the 20% Na electrode was limited to lower current densities than composites with less Na. Further experiments were carried out at zero applied stack pressure (fig. S6 and supplementary text 1). Data at different current densities were fit to Sand's equation to extract an interfacial transport descriptor (*23*, *34*), revealing that increasing Na content in the electrodes systematically enhances the effective transport of Li to the SSE interface. This implies that the incorporated Na phase stabilizes interfacial contact at zero stack pressure.

Electrochemical behavior during further plating and stripping was also investigated (fig. S5). After initial stripping, 3 mA·hour cm^{-2} of Li was redeposited onto the electrodes, followed by a second stripping process. Pure Li electrodes short circuited under all current density/stack pressure conditions during redeposition. Many of the Li-Na electrodes, and particularly those with 5 and 10% Na, did not short circuit upon plating (fig. S5). During the second stripping, the composite electrodes tended to exhibit higher polarization than during the first, but stable stripping to 3 mA·hour cm^{-2} was achieved for the composite electrodes across a range of current densities and stack pressures. Electrodes with 10 and 20% Na showed the lowest voltage at 0.8 MPa stack pressure during the second stripping. These results show that the presence of Na contributes to stable electrodeposition and further stripping steps.

Interfacial evolution

To provide insight into interfacial evolution, we conducted potentiostatic electrochemical impedance spectroscopy (EIS) during Li stripping. We show in Fig. 2, A and B, galvanostatic stripping curves and Nyquist plots collected at increments of 0.5 mA·hour cm^{-2} capacity from pure Li and 10% Na electrodes at 1.6 MPa stack pressure. The spectral features at high frequency (~1 MHz) likely arise from bulk SSE resistance, whereas lower frequency (100 to 1 kHz) features are associated with interfacial resistance, including interphase and charge transfer resistance (*5*, *25*, *26*, *35*). At the onset of stripping, the voltages of the pure Li cell (Fig. 2A) and the 10% Na cell (Fig. 2B) were both 11 mV. The initial EIS spectra of both cells were also quite similar, showing compact spectra with a partial semicircle in the high-frequency

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range (>1 kHz). Their minima along the real axis, representing interfacial resistance, were observed at 36 to 38 ohm cm^2. After stripping of the pure Li electrode to 5 mA·hour cm^{-2}, the cell voltage increased to 63 mV, and the high-frequency semicircle in the impedance spectrum expanded, representing an increase in interfacial resistance to ~200 ohm cm^2 (Fig. 2A). Additionally, the apex frequency of this semicircle shifted to lower values during stripping (fig. S7A), indicating that this feature is associated with voids at the interface (*26*, *36*, *37*). By contrast, the impedance spectrum of the 10% Na cell (Fig. 2B) showed no substantial increase in its partial semicircle and only a minor shift of interfacial resistance to ~41 ohm cm^2 during stripping, which likely includes the impact of interphase growth. This behavior was accompanied by a smaller voltage increase to 41 mV in the galvanostatic data. These EIS spectra showed the formation of a low-frequency tail (<1 kHz), which was also observed in Li-Na electrodes with other compositions (fig. S8). Higher Na concentrations typically featured earlier tail development and higher voltages in the galvanostatic profiles at the end of stripping (figs. S7, B and C, and S8).

The evolution of the Li-Na EIS spectra allows for the following conclusions. The lack of semicircle evolution indicates that the electrode-SSE interface retains low resistance with minimal void formation during stripping, and the growth of the low-frequency tails is likely due to interfacial accumulation of the ion-blocking Na phase (fig. S9 describes control experiments with Na||Na symmetric cells demonstrating blocking behavior) (*38*, *39*). This result implies that excessive Na content (20% Na and above) causes faster voltage polarization because of the more rapid interfacial accumulation of Na to block Li transport, rather than through void formation. The interfacial reaction between the accumulated Na and the SSE can be effectively controlled during operation by ensuring that the electrode potential remains below ~0.3 V versus Li/Li^+, at which Na reactions contribute to interphase formation (figs. S4 and S10). Such control maintains low interfacial impedance and minimal interphase formation (fig. S10).

Morphological evolution was investigated with cryo-FIB-SEM (Fig. 2, E and F). In the initial state, a 10% Na electrode (Fig. 2E and fig. S11A) shows Na domains distributed uniformly throughout the darker Li, without contact between Na and the SSE material. After stripping 4 mA·hour cm^{-2} in a different cell (Fig. 2F), the electrode showed a few-micrometer-thick Na layer that contacted the SSE. This indicates accumulation of Na at the interface during Li stripping, with the Na maintaining electrical contact (Fig. 2B). When Li was fully stripped from a 10% Na electrode, Na remained at the SSE interface with only minor residual Li present, supporting 74% Li utilization from the composite electrode (fig. S12).

The reversibility of this behavior was investigated to understand how the accumulated Na interfacial layer evolves upon subsequent Li plating. As displayed in Fig. 2C, stable galvanostatic Li deposition occurs without nucleation overpotential after the first stripping. We provide in Fig. 2G a cryo-FIB-SEM image that demonstrates the accumulated Na layer shifting away from the interface as Li is deposited onto it. The stripping steps during the second and third cycles showed EIS evolution and final voltage values similar to those of the first

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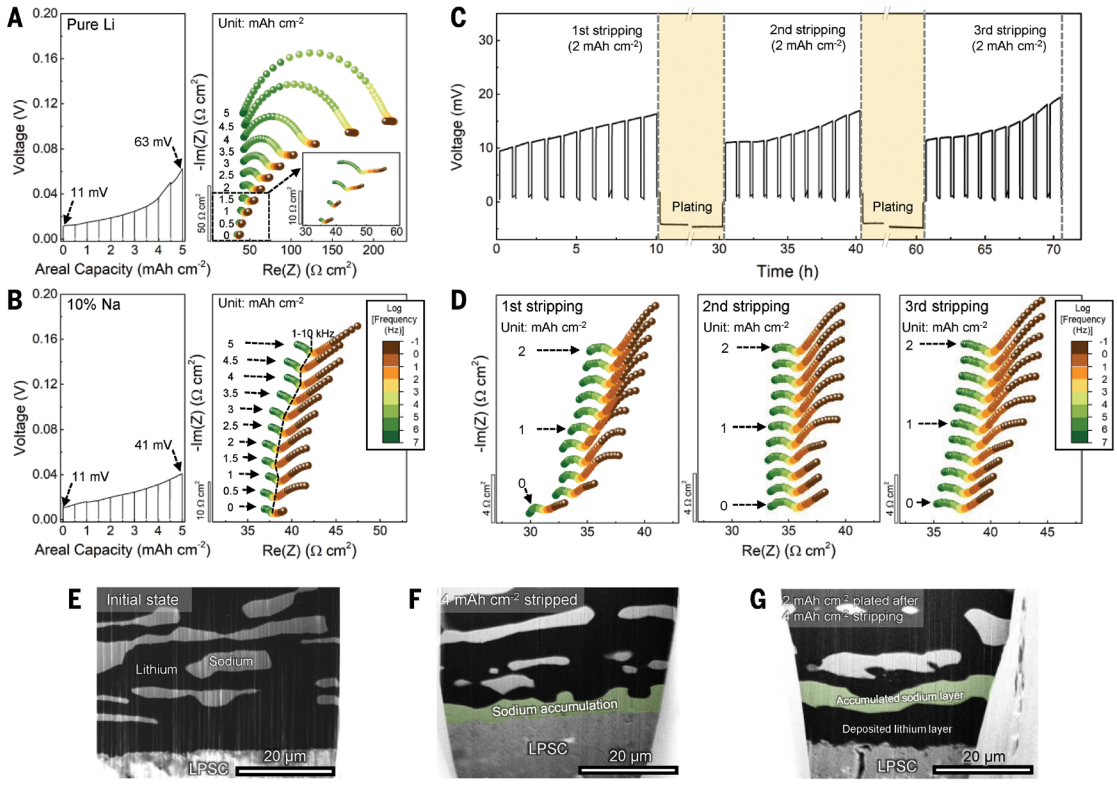


Fig. 2. Electrochemical impedance analysis of interfacial evolution. (**A** and **B**) In situ EIS analysis of (A) the pure Li and (B) 10% Na electrodes during Li stripping with Li counter electrodes at 1.6 MPa stack pressure. (Left) Voltage profiles. (Right) The associated Nyquist plots. A current density of 0.25 mA cm^{-2} was used with potentiostatic EIS spectra taken every 0.5 mA·hour cm^{-2}, as marked. Impedance spectra for other compositions are shown in fig. S7. (**C** and **D**) In situ EIS analysis of a 10% Na electrode during electrochemical cycling with a Li counter electrode at 2.5 MPa stack pressure. (C) Voltage profile and (D) associated Nyquist plots for the first, second, and third stripping steps. Current densities of 0.25 mA cm^{-2} and 0.1 mA cm^{-2} were used for stripping and plating, respectively. Potentiostatic EIS spectra were measured every 0.2 mA·hour cm^{-2} during stripping, as marked. (**E** to **G**) Cryo-FIB-SEM images of 10% Na electrodes at various stages of cycling: (E) the initial state, (F) after 4 mA·hour cm^{-2} was stripped, and (G) after 2 mA·hour cm^{-2} was plated (after initial stripping of 4 mA·hour cm^{-2}). All cells in (E) to (G) had a stack pressure of 1.6 MPa and current density of 0.25 mA cm^{-2}. In (F) and (G), an artificial green color has been overlaid on interface-accumulated Na. Images without the overlays are provided in fig. S11, B and C.

stripping step (Fig. 2, C and D). The low-frequency EIS tails appear during these stripping steps similarly to the first step. This is consistent with a cryo-FIB-SEM image in fig. S11D, which shows interfacial Na accumulation after partial stripping after multiple cycles. These results indicate that Na returns to contact the SSE interface during cyclic Li stripping without substantial void formation. Moreover, the EIS signatures suggest that Na accumulates at the interface even before the deposited Li layer is fully depleted (fig. S11D). Li deposition onto the accumulated Na layer (Fig. 2G) also implies that a Na metal layer alone could serve as an effective current collector for Li cycling. This feasibility was further validated in fig. S13, where Na metal as a current collector demonstrated extended cycling capability compared with that of a Cu current collector; however, higher Coulombic efficiency was attained with composite Li-Na electrodes.

Operando x-ray computed tomography

We investigated the dynamic evolution of a Li-Na electrode using operando synchrotron x-ray computed tomography (XCT) imaging. A half cell with a 10% Na electrode was used in a cell customized for XCT during electrochemical testing (supplementary materials, materials and methods). A cross-sectional image slice of the cell stack is shown in Fig. 3A. The grayscale contrast between both electrodes and LPSC is due to differences of the x-ray attenuation coefficients (*8*, *40*), with the electrodes appearing darker because they are primarily composed of low-Z Li metal (*8*, *41*).

Galvanostatic stripping and plating were performed sequentially by using a current density of 0.5 mA cm^{-2}, with stripping and plating capacities of 1.0 and 1.13 mA·hour cm^{-2}, respectively (fig. S14). Magnified image slices of a location at the interface between the SSE and the Li-Na electrode at various times are shown in Fig. 3B. The bright domains in the electrode are Na. To highlight Na evolution, grayscale intensity profiles across the interface are displayed below the image slices in Fig. 3B, with scanned locations indicated with red boxes. The electrode-SSE interface is indicated with a red line in each intensity plot. As shown in the profile in Fig. 3B, i, the composite electrode exhibits greater intensity variation compared with that in the SSE region. Grayscale intensity peaks and valleys in the electrode correspond to the Na domains and the Li matrix, respectively. The average grayscale pixel intensity values of the electrode and the SSE regions in this profile are 98.1 ± 65.2 and 124.3 ± 22.9, respectively, which is consistent with other locations (fig. S15 and supplementary text 3).

Three different Na domains were tracked during electrochemical cycling, with the peaks associated with these domains indicated

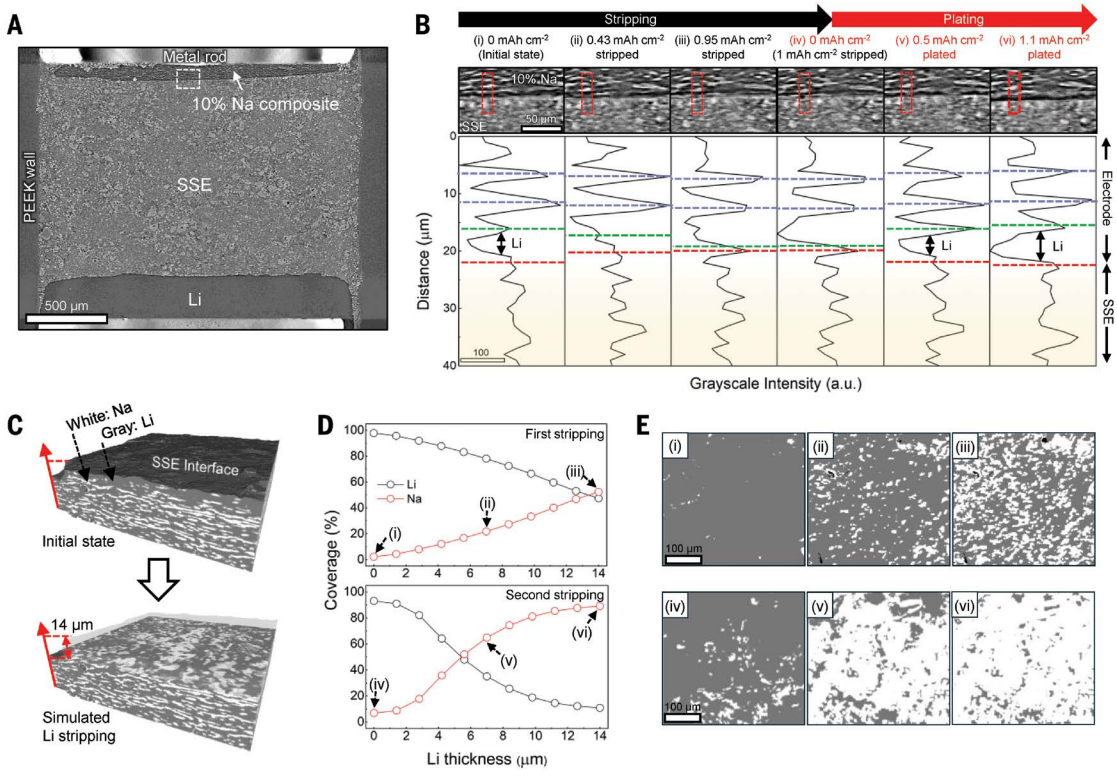


Fig. 3. Operando x-ray CT analysis of interfacial evolution. (**A** and **B**) Operando x-ray tomography imaging of a 10% Na composite electrode half cell during Li stripping and plating. (A) An image slice of the 10% Na electrode|LPSC|Li half cell. (B) Cross-sectional images of a portion of the interface between LPSC and the Li-Na electrode at different stages of cycling, along with corresponding grayscale intensity profiles from the red boxes in the images. The stages of electrochemical cycling are labeled from left to right as follows: (i) 0 mA·hour cm^{-2} stripped (initial state), (ii) 0.43 mA·hour cm^{-2} stripped, (iii) 0.95 mA·hour cm^{-2} stripped, (iv) 0 mA·hour cm^{-2} plated (equivalent to 1 mA·hour cm^{-2} stripped), (v) 0.5 mA·hour cm^{-2} plated, and (vi) 1.1 mA·hour cm^{-2} plated. The electrochemical cycling profile is shown in fig. S14. In the grayscale profiles, the SSE interface is indicated with red dashed lines. The nearest Na domain (high grayscale intensity) to the interface is indicated with a green dashed line, and other Na domains are indicated with blue dashed lines. The white box in (A) indicates the location of these cross-sectional images. (**C** to **E**) Simulated stripping analysis to determine the evolution of Li and Na coverage at the SSE interface. (C) Rendered 3D subvolumes (top) in the pristine state and (bottom) after simulated Li stripping, where Li is gray and Na is white. The reconstructed initial volume measures 420 by 420 by 131 µm. The accumulation of Na at the SSE interface was simulated by removing Li voxels from the initial state to a thickness of 14 µm (10 voxel layers), while Na remained (fig. S16 and materials and methods). (D) Interfacial coverage of Na and Li in the subvolume during the (top) first and (bottom) second stripping processes by means of simulated Li stripping. (E) 2D interfacial coverage maps at different simulated stripping states, as marked in (D). Li is gray and Na is white.

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in Fig. 3B with green and blue dashed lines in the intensity profiles. The Na domain closest to the SSE interface (Fig. 3B, green dashed line) initially remained separated from the SSE interface (Fig. 3B, red dashed line) by a Li layer (Fig. 3B, i). During Li stripping, this Na peak moved closer and became conflated with the SSE interface (Fig. 3B, ii to iv). During Li plating, the Na peak again became distinct as Li deposited below it (Fig. 3B, iv to vi). This behavior is consistent with the ex situ SEM in Fig. 2G and demonstrates the accumulation and removal of Na from the interface. The other Na peaks in the profile were also displaced as Li was stripped and redeposited (Fig. 3B, blue dashed lines). This Na domain redistribution was consistent with analysis of a different location (fig. S15).

As previously discussed, the interfacial distribution of Na is different after one cycle compared with its initial state. To understand how the interfacial accumulation of Na depends on the initial distribution of Na within the microstructure, we developed an analysis to simulate Li stripping using the three-dimensional (3D) XCT data. This "simulated stripping" analysis was carried out instead of direct segmentation of Na because the Na and the SSE phases exhibited similar grayscale intensity (Fig. 3B), which precluded accurate segmentation of contacting phases. The reconstructed microstructure of the pristine composite electrode is shown in Fig. 3C. The simulated stripping process involved virtually "removing" a planar layer of Li voxels and projecting the Na voxels in the same layer onto the SSE interface, followed by identical processing of the next layers (fig. S16). The evolution of interfacial coverage of Li and Na is shown in Fig. 3D as a function of stripped Li thickness up to 14 μm (10 voxel layers, or 2.9 mA·hour cm^{-2}). Stripping was simulated for both the pristine XCT microstructure (Fig. 3B, i) and the microstructure after one cycle (Fig. 3B, vi). The corresponding 2D maps of Li and Na coverage across a region of the interface are shown in Fig. 3E. As shown in Fig. 3, D and E, i to iii, the interfacial coverage of Na during the first stripping increased steadily as Li was stripped and its coverage decreased. During the second stripping, however, a more abrupt increase of Na coverage occurred (Fig. 3, D and E, iv to vi). This more abrupt increase is due to the deposited Li layer beneath the accumulated Na being depleted to result in sudden Na-SSE contact. Thus, the microstructural distribution of Na is expected to strongly influence Na contact pattern formation at the interface and resulting electrochemical behavior.

Electrochemical cycling

We carried out galvanostatic cycling tests to investigate electrochemical reversibility. Cells with Li–In (indium) counter electrodes were cycled with 2 mA·hour cm^{-2} capacity per cycle. All the Li-Na electrodes with different compositions showed reversible cycling over 10 cycles (Fig. 4A). Because the Li-In counter electrode maintains a constant electrode potential during lithiation-delithiation (+0.62 V versus Li/Li^+) (*28*), the voltage profiles in Fig. 4A primarily reflect the behavior of the Li-Na electrodes. The overlaid stripping profiles over four cycles are shown in Fig. 4B. The voltage profiles during the first stripping generally exhibited a linear increase, with their slopes approximately proportional to Na concentration. Subsequent stripping profiles showed nonlinear increases, especially with higher Na concentrations. The Na distribution therefore affects the extent of voltage polarization because the more abrupt accumulation of the Na phase at the SSE interface observed in the second cycle in the XCT data in Fig. 3D gives rise to more rapid polarization. The 5 and 10% Na electrodes also exhibited stable cycling upon doubling the cycling capacity to 4 mA·hour cm^{-2} per cycle (fig. S17).

We also conducted longer-term galvanostatic cycling tests. We show in Fig. 4, C and D, cycling of a 5% Na electrode at a current density of 0.75 mA cm^{-2} and a stack pressure of 2.5 MPa (Fig. 4C) and a 10% Na electrode at a current density of 0.5 mA cm^{-2} and a stack pressure of 1.6 MPa (Fig. 4D). Under these conditions, pure Li electrodes typically exhibited rapid voltage polarization during stripping owing to interfacial voiding, followed by short circuiting during plating (Fig. 4, C and D, insets). By contrast, the cells with Li-Na electrodes showed substantially enhanced cyclability and low voltage. The 5% Na electrode (Fig. 4C) and the 10% Na

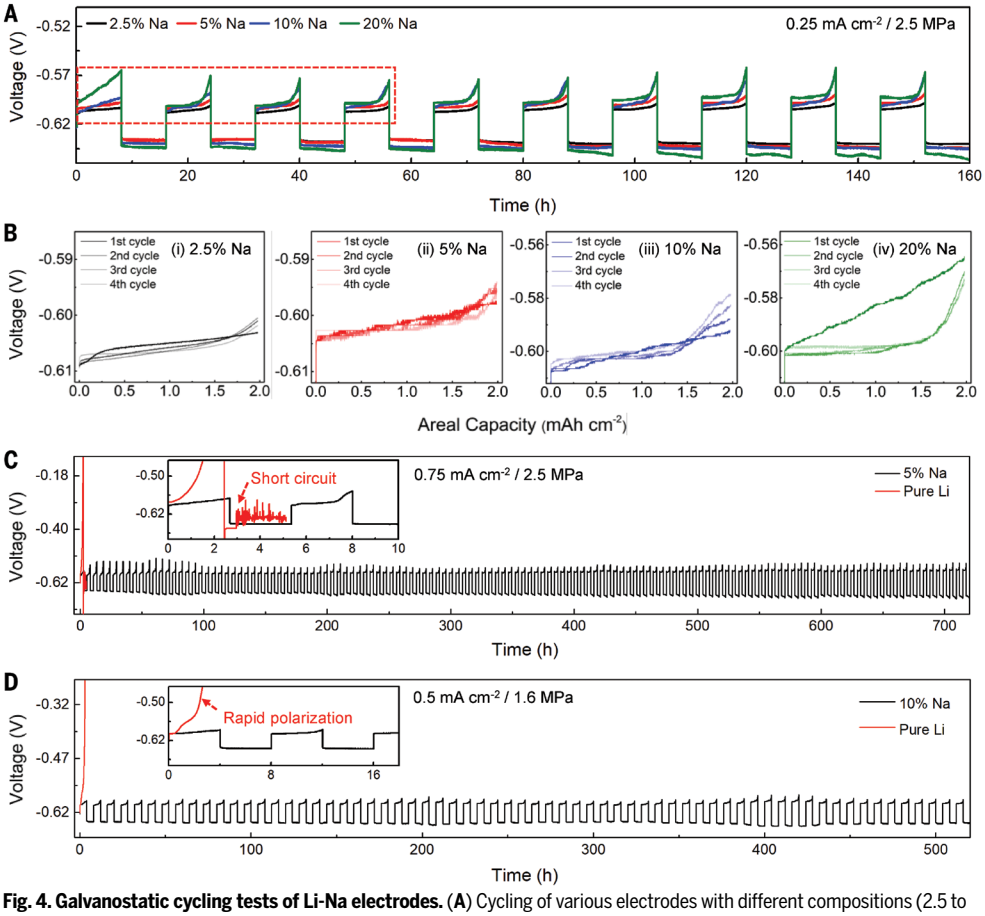


Fig. 4. Galvanostatic cycling tests of Li-Na electrodes. (**A**) Cycling of various electrodes with different compositions (2.5 to 20 atomic % Na) at a current density of 0.25 mA cm^{-2} and a stack pressure of 2.5 MPa. (**B**) Corresponding stripping voltage profiles for the first four cycles, as highlighted within the red box in (A). (**C** and **D**) Long-term cycling stability of (C) 5% Na and (D) 10% Na electrodes. For comparison, pure Li electrodes were cycled under the same conditions as in (C) and (D). The cells in (C) were cycled at a current density of 0.75 mA cm^{-2} and a stack pressure of 2.5 MPa, and those in (D) were cycled at a current density of 0.5 mA cm^{-2} and a stack pressure of 1.6 MPa. All cells used a Li–In counter electrode and were tested with an areal capacity of 2 mA·hour cm^{-2} per cycle. The measured voltage values displayed are negative because the Li–In counter electrode exhibits an electrode potential of 0.62 V versus Li/Li^+.

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electrode (Fig. 4D) sustained over 100 and 50 cycles, respectively. Additionally, the 5% Na electrode demonstrated outstanding cyclability at a current density of 1 mA cm^{-2} and a stack pressure of 4.1 MPa (fig. S18). These results demonstrate that the accumulation of Na at the SSE interface not only mitigates voiding, but because of its electrochemical inactivity and deformability, it also supports stable Li plating (Fig. 2G), enabling cycling at low stack pressures.

All-solid-state full cells with Li-Na anodes and sulfur cathodes were fabricated and tested under low stack pressures between 2.5 and 0.8 MPa (figs. S19 and S20 and supplementary text 4), which is much lower than most literature demonstrations of sulfur SSBs (*42*, *43*). A cell with a 5% Na anode displayed >3 mA·hour cm^{-2} and 75% capacity retention after 100 cycles with 2.5 MPa stack pressure (fig. S19), whereas cells with pure Li anodes short circuited in approximately five cycles. A cell with a 10% Na anode exhibited 69 cycles at a stack pressure of 0.8 MPa (fig. S20A), whereas a pure Li cell short circuited in the first cycle.

In fig. S2, we compare the experimental conditions (stack pressures and current density) from the results herein with other published SSB demonstrations that used Li metal and alloy electrodes with sulfide SSEs. Alloy anodes such as Si are frequently used to overcome the challenges of Li metal (*27*–*29*). However, both Li metal and Li alloy anodes usually necessitate high stack pressures, ranging from a few megapascals to tens of megapascals (fig. S2). The required stack pressure generally rises with increased current density, with higher stack pressures needed to maintain the morphological conformity of the solid-solid interface (*29*, *44*). Our multiphase alkali metal electrodes deviate from this trend, functioning effectively at stack pressures between 0.8 and 2 MPa, with modest current densities (fig. S20). Incorporating denser Na into the Li anode reduces specific energy by only 1 to 2% (table S5 and fig. S21), allowing for mechanistic benefits to be realized with negligible impact on energy metrics.

Electro-chemo-mechanical modeling

We developed a mesoscale model to better understand how electro-chemo-mechanical interactions influence Na domain evolution at the SSE interface. The model considers electrochemical reaction kinetics, mechanical deformation of the metals, diffusivity, and ionic transport (supplementary materials). The stress distribution in a pure Li electrode near a void at the Li-SSE interface is illustrated in Fig. 5A (1 MPa stack pressure). The stresses diminish above the void because of the absence of Li/SSE contact, whereas stress concentration in the Li occurs near the void edges. Current focusing near the void edges during stripping leads to contact loss at the Li-SSE interface and void growth (Fig. 5, B and C). By contrast, a Na domain at the interface contacts both the SSE and Li, homogenizing the stress distribution with the Na domain under compressive stress (Fig. 5D). Because Na has a lower yield strength than that of Li (nanoindentation shows 54% lower hardness) (fig. S23) (*45*–*47*), the stress exerted on the Na domain by the surrounding Li can induce plastic deformation and creep along the interface as Li is stripped. The simulation shows that this causes Na to deform and extend along the SSE interface as Li is removed (Fig. 5, E and F), suppressing the formation of voids. From these results, we expect that it is the chemo-mechanical behavior of Na that enhances stripping behavior rather than a pure transport effect; density functional theory calculations showed that Li-Na interfaces exhibit lower diffusivity than do Li surfaces (fig. S24 and table S7).

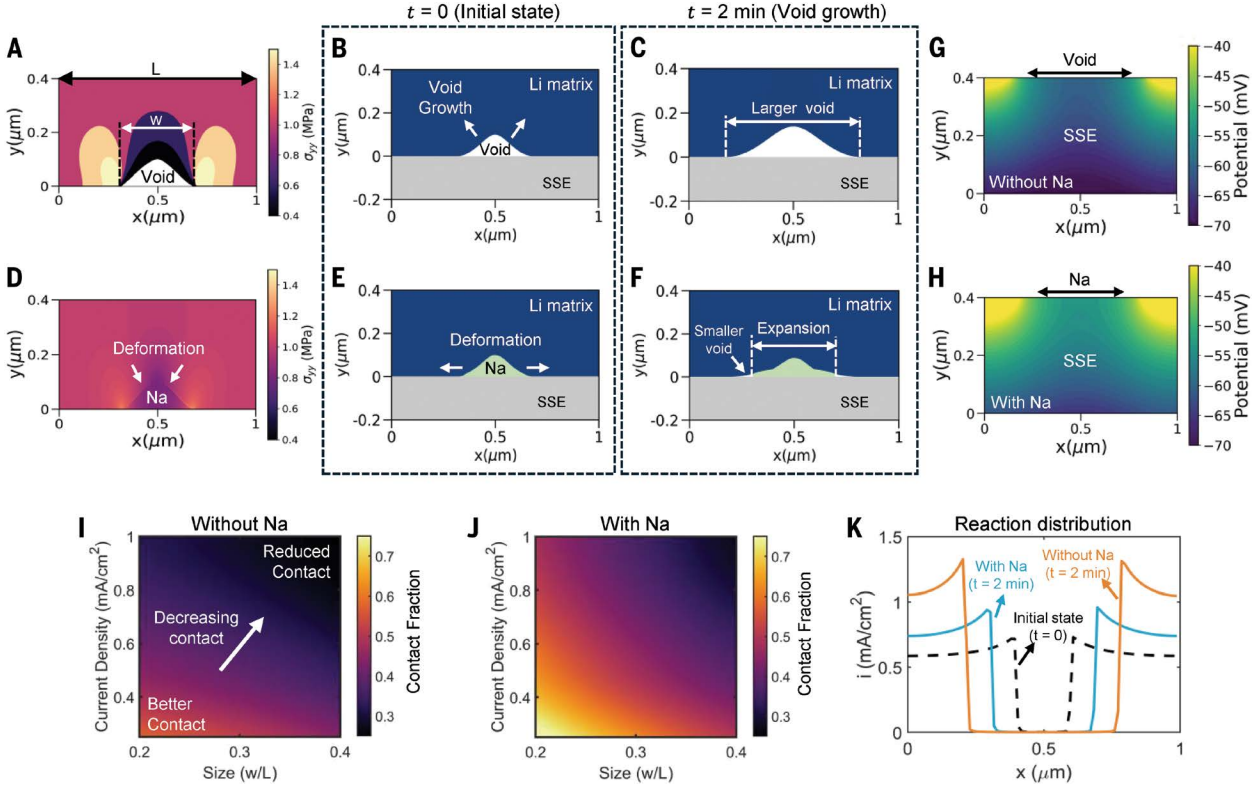


Fig. 5. Electro-chemo-mechanical modeling of the evolution of the SSE interface. (**A**) Normal stress along the *y* direction in Li metal with a void at the SSE interface. The void size $w/L = 0.3$ (where w is the void width and L is the model dimension). (**B** and **C**) Li-SSE interfacial evolution and void growth from time (B) $t = 0$ to (C) $t = 2$ min, with an applied current density (I_{App}) of 0.5 mA cm^{-2}. (**D**) Normal stress along the *y* direction in Li metal with Na at the SSE interface (Na size, $w/L = 0.3$). The other stress components are included in fig. S22. (**E** and **F**) Li-SSE interfacial evolution from time (E) $t = 0$ to (F) $t = 2$ min, when Na is present at the interface ($I_{App} = 0.5$ mA cm^{-2}). (**G** and **H**) Electric potential distribution in the SSE with (G) void and (H) Na at the Li-SSE interface. (**I** and **J**) Effects of current density and void or Na size on Li-SSE contact fraction with (I) void and (J) Na domain. (**K**) Reaction current density at the Li-SSE interface with and without Na at time $t = 0$ and $t = 2$ min. I_{App} is 0.5 mA cm^{-2}, and the void or Na domain size (w/L) is 0.2. The stack pressure applied in this model is 1 MPa. The mathematical formulation used for this model is described in supplementary text 7.

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The electric potential distributions within the SSE after 2 min of stripping for both simulated cases are shown in Fig. 5, G and H, for an applied current density $I_{App} = 0.5$ mA cm^{-2}. The greater Li-SSE contact achieved with Na results in less current focusing at the interface (fig. S25) and a more uniform potential distribution in the SSE domain. The effect of void or Na domain size and applied current density on Li-SSE contact evolution at time (t) = 2 min is shown in Fig. 5, I and J. The reaction heterogeneity at the interface increases and Li-SSE contact decreases with an increase in applied current density. An increase in void or Na domain size also exacerbates current heterogeneity owing to the reduced Li-SSE interfacial contact. However, the action of Na in suppressing voids sustains higher interfacial contact fractions at higher current densities and domain sizes. The difference in Li-SSE contact evolution with and without Na is more pronounced for smaller voids because of the rapid growth of the void through Li surface diffusion. In Fig. 5K, we compare the reaction distribution for a void versus Na domain with $I_{App} = 0.5$ mA cm^{-2} in the initial state ($t = 0$ min) and after $t = 2$ min. Initially, the interfacial contact region for the void and Na are the same, resulting in an identical reaction current distribution (Fig. 5K). As the interface evolves over time, more contact loss and current focusing occurs near the void compared with Na. Overall, these simulation results show that the deformability of Na promotes dynamic interfacial contact in response to local chemo-mechanical stimuli, suppressing void formation as Na accumulates at the interface.

Conclusions

We have demonstrated the concept of interfacial morphogenesis applied to Li metal SSBs, in which interfacial properties are self-regulated across length scales in a multiphase electrode through global contact pattern formation promoted by local chemo-mechanical stimuli. Binary Li-Na electrodes show the capability to suppress void growth through accumulation of highly deformable Na domains at the SSE interface, enhancing Li stripping capacity and cycling at low stack pressures (<3.2 MPa). The electrochemical inertness of the Na phase with respect to active Li enables the Na to avoid the structural degradation often seen in Li-reactive metal alloy or carbon interlayers and composites (*25*, *29*). Additionally, binary alkali metal foils can be created and cold-rolled with simple metallurgical techniques, avoiding more complex processing (*6*, *23*, *24*, *26*). Overall, this study shows the promise of engineering Li metal electrodes with tailored chemo-mechanical properties to promote desired morphology evolution. To address the diverse demands of battery applications (such as in electric vehicles), it is essential to investigate the chemo-mechanical evolution of materials under varying conditions, such as different charging-discharging rates, temperatures, and charging modes (such as current pulsing) (*9*, *48*, *49*). Such investigations could enable multicomponent metal anodes to achieve enhanced performance metrics that would be attractive for practical applications.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Supplementary Text; Figs. S1 to S25; Tables S1 to S8; References (*50*–*100*); Data S1 and S2

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Interface morphogenesis with a deformable secondary phase in solid-state lithium batteries

Sun Geun Yoon, Bairav S. Vishnugopi, Douglas Lars Nelson, Adrian Xiao Bin Yong, Yingjin Wang, Stephanie Elizabeth Sandoval, Talia A. Thomas, Kelsey Anne Cavallaro, Pavel Shevchenko, Elif P#nar Alsaç, Congcheng Wang, Aditya Singla, Julia R. Greer, Elif Ertekin, Partha P. Mukherjee, and Matthew T. McDowell

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Editor's summary

A challenge for lithium solid-state batteries is the formation of voids at the lithium-electrolyte interface as the battery is cycled because the loss of interfacial contact degrades battery performance. The typical solution to this problem is to apply a stack pressure to deform the lithium, but the pressures required become unfeasible. Yoon *et al.* eliminated the need for high pressure by alloying the lithium with up to 20% sodium (see the Perspective by Spencer-Jolly). Because the sodium is immiscible, it forms domains within the lithium microstructure. However, it is also a good conductor, so the accumulation at the solid electrolyte interface ensures good electrical contact but it will still move away from the interface as the lithium plates out during charging. —Marc S. Lavine

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